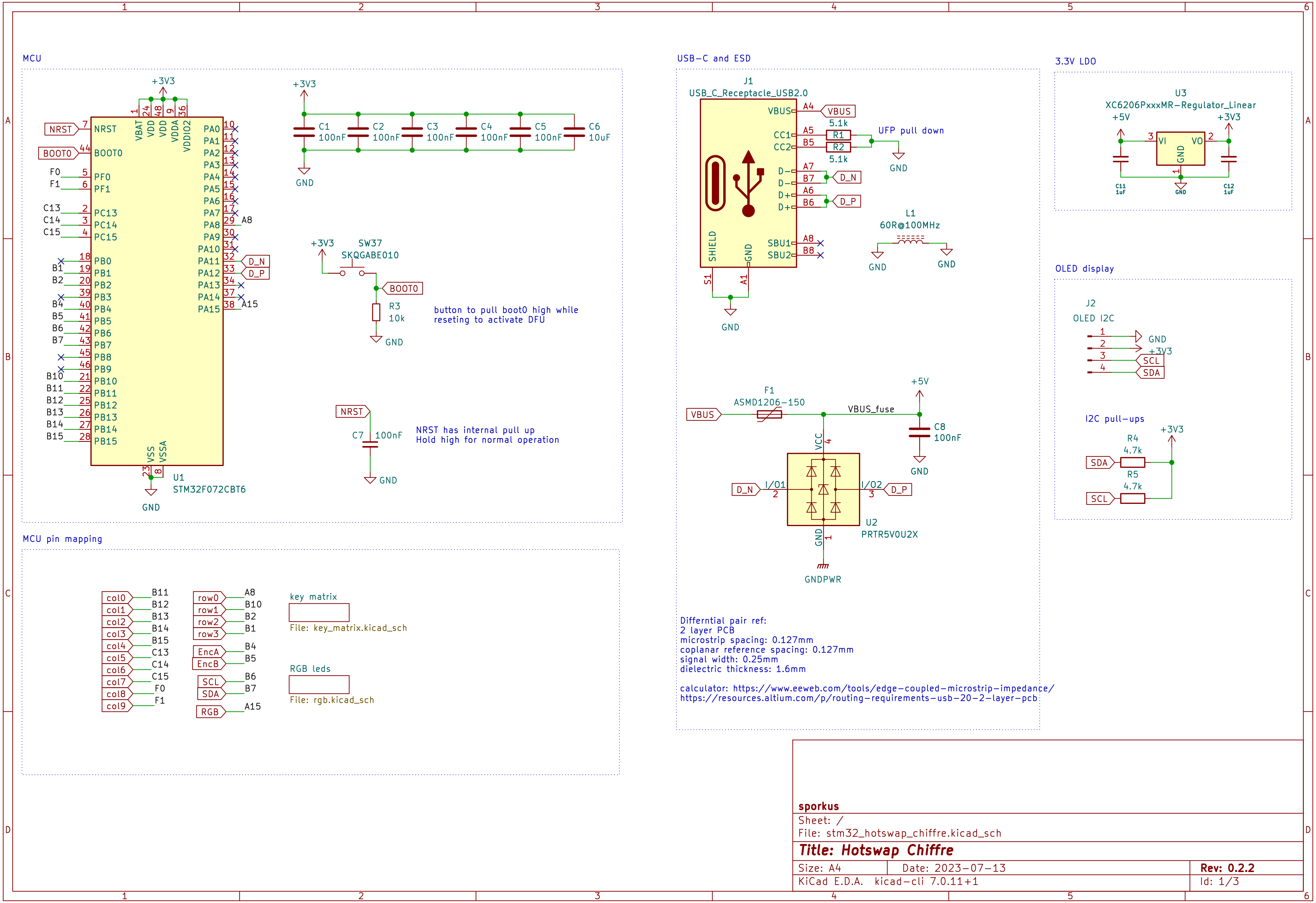
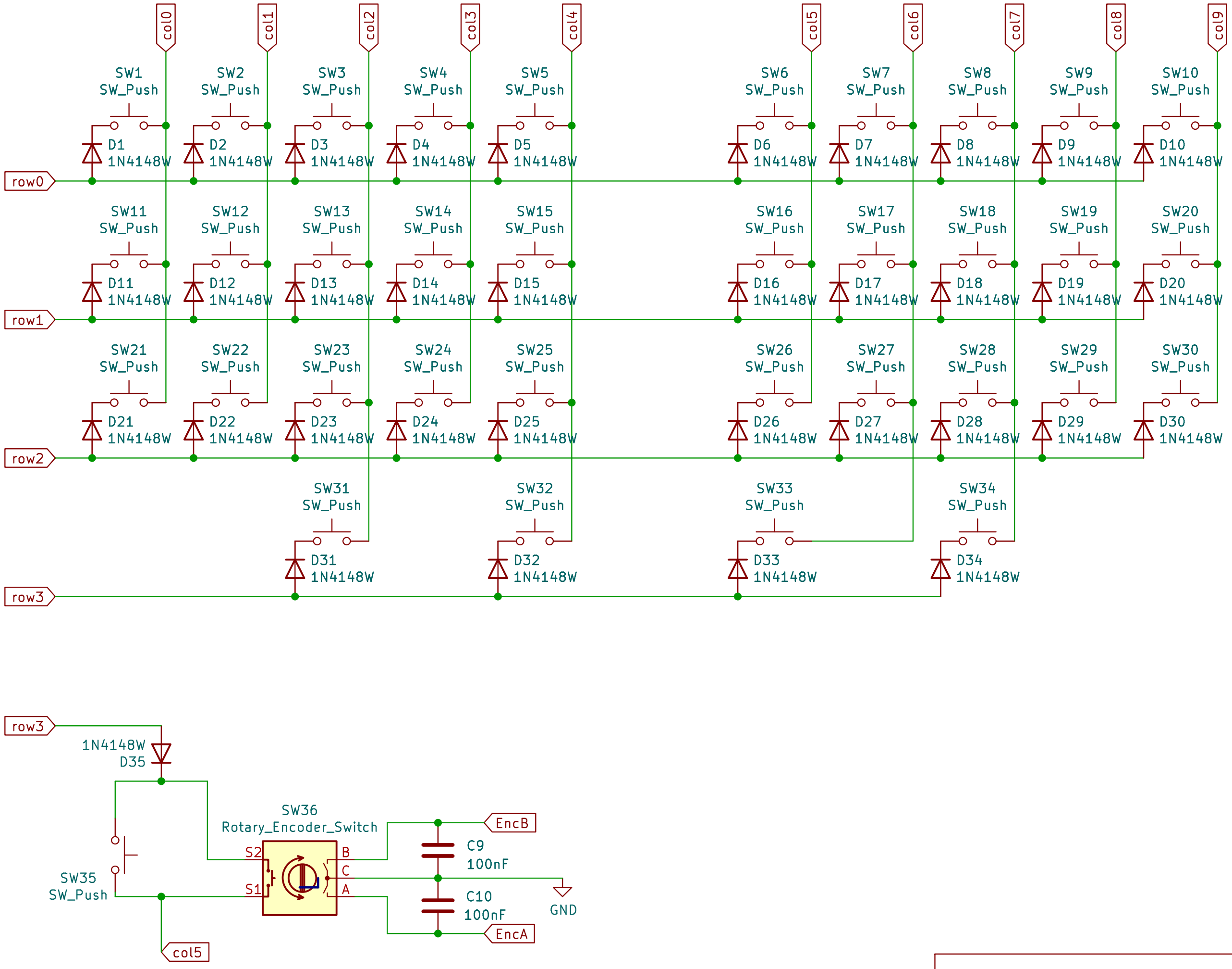

MCU
+3V3
NRST
BOOT0
F0
F1
C13
C14
C15
B1
B2
B4
B5
B6
B7
B10
B11
B12
B13
B14
B15
NRST BOOT0 PF0 PF1 PC13 PC14 PC15 PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB9 PB10 PB11 PB12 PB13 PB14 PB15
VBAT VDD VDD VDDA VDDIO2
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
A8
D_N
D_P
A15
VSS VSSA
U1
STM32F072CBT6
GND
C1 100nF
C2 100nF
C3 100nF
C4 100nF
C5 100nF
C6 10uF
SW37
SKQGABE010
R3
10k
button to pull boot0 high while
reseting to activate DFU
C7 100nF
NRST has internal pull up
Hold high for normal operation
MCU pin mapping
col0 B11
col1 B12
col2 B13
col3 B14
col4 B15
col5 C13
col6 C14
col7 C15
col8 F0
col9 F1
row0 A8
row1 B10
row2 B2
row3 B1
EncA B4
EncB B5
SCL B6
SDA B7
RGB A15
key matrix
File: key_matrix.kicad_sch
RGB leds
File: rgb.kicad_sch
USB-C and ESD
J1
USB_C_Receptacle_USB2.0
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
GND
R1 5.1k
R2 5.1k
UFP pull down
L1
60R@100MHz
F1
ASMD1206-150
VBUS_fuse
+5V
C8
100nF
VCC
I/O1
I/O2
U2
PRTR5V0U2X
GNDPWR
Differntial pair ref:
2 layer PCB
microstrip spacing: 0.127mm
coplanar reference spacing: 0.127mm
signal width: 0.25mm
dielectric thickness: 1.6mm
calculator: https://www.eeweb.com/tools/edge-coupled-microstrip-impedance/
https://resources.altium.com/p/routing-requirements-usb-20-2-layer-pcb
3.3V LDO
U3
XC6206PxxxMR-Regulator_Linear
+5V
+3V3
VI
VO
C11 1uF
C12 1uF
OLED display
J2
OLED I2C
GND
+3V3
SCL
SDA
I2C pull-ups
R4 4.7k
R5 4.7k
sporkus
Sheet: /
File: stm32_hotswap_chiffre.kicad_sch
Title: Hotswap Chiffre
Size: A4
Date: 2023-07-13
Rev: 0.2.2
KiCad E.D.A. kicad-cli 7.0.11+1
Id: 1/3



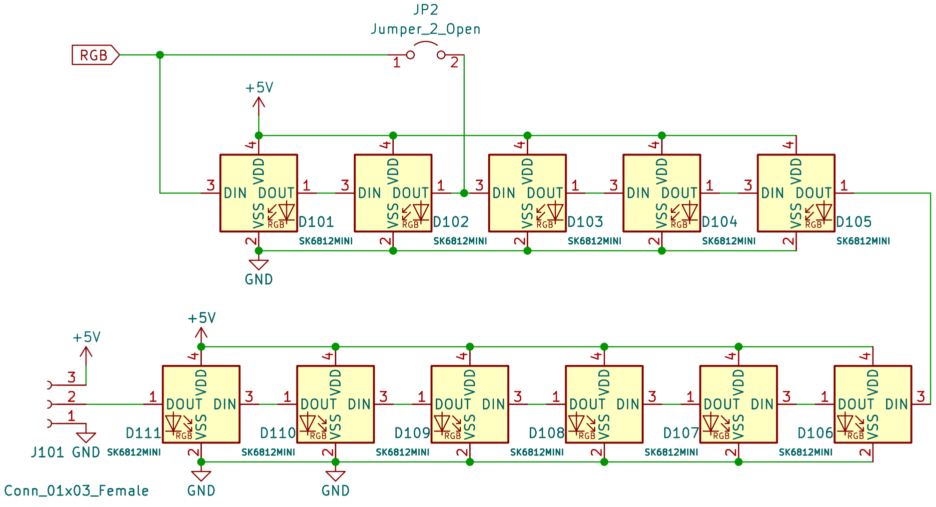
sporkus

Sheet: /key matrix/

File: key_matrix.kicad_sch

Title: Hotswap Chiffre

Size: A4	Date: 2023-07-13	**Rev: 0.2.2**
KiCad E.D.A. kicad-cli 7.0.11+1		Id: 2/3



sporkus

Sheet: /RGB leds/

File: rgb.kicad_sch

Title: Hotswap Chiffre

Size: A4	Date: 2023-07-13	**Rev: 0.2.2**
KiCad E.D.A. kicad-cli 7.0.11+1		Id: 3/3